R26-IT-010

AI-Based Electricity Demand Intelligence System

for Behavior, Prediction, and Decision Support



Thanujan T | IT22545176 — Behaviour Intelligence

Janahan R | IT22069740 — Adaptive Forecasting

Madushani R.P | IT22273062 — Explainable & Causal AI

Dewmini D.M.D.P | IT22630216 — Scenario & Decision Support

Research Team & Component Overview

Each member develops one AI module — together forming a unified Electricity Demand Intelligence Platform

🔍 Component 01

Thanujan T

IT22545176

Behaviour Intelligence

K-Means · DBSCAN · Isolation Forest · Behaviour Labels

📈 Component 02

Janahan R

IT22069740

Adaptive Forecasting

CNN-LSTM · Transformer · Probabilistic Confidence Intervals

🧠 Component 03

Madushani R.P

IT22273062

Explainable & Causal AI

SHAP · LIME · Causal Factor Analysis · XAI Reports

⚙️ Component 04

Dewmini D.M.D.P

IT22630216

Scenario & Decision Support

What-If Simulation · Risk Assessment · Alert Generation

Research Gap

What existing tools offer — and the critical gaps that justify this research

✓ What Currently Exists	✗ Critical Gaps Identified
• ARIMA/SARIMA/LSTM — predict numbers, no explanation	→ No day-level behaviour labeling — operators work blind
• EnergyHub, OpenEnergyMonitor — dashboards & basic alerts	→ No uncertainty quantification — point forecasts only
• Anomaly detection — only at individual meter/fraud level	→ No interpretable causal explanations for demand changes
• Clustering — used as preprocessing only, not insights	→ No what-if scenario simulation or risk planning tool
• XAI research — exists in isolation, not integrated	→ No unified platform: all 4 capabilities in one system

Our Solution — 4 Integrated AI Modules

A unified platform that understands, predicts, explains, and plans electricity demand

🔍 01

Behaviour Intelligence

Classifies daily demand into patterns, detects anomalies, labels each day

📈 02

Adaptive Forecasting

CNN-LSTM + Transformer probabilistic forecasting with confidence ranges

🧠 03

Explainable & Causal AI

SHAP/LIME reveals WHY demand changes — weather, holidays, industry

⚙️ 04

Scenario Decision Support

Simulates what-if scenarios, assesses risk levels, generates alerts

Result: Electricity planners know WHAT demand will be · WHY it changes · HOW it behaves · WHAT IF conditions change

System Architecture

5-layer platform — Client → Frontend → Backend → AI Layer → Data

CLIENT	Web Browser / Device — Energy planners · CEB operators · Grid analysts
FRONTEND	React.js Dashboard — Behaviour labels · Forecast charts · XAI reports · Scenario sliders
API GATEWAY	FastAPI — Authentication · Rate limiting · HTTPS routing · JWT security
AI LAYER	Behaviour Intelligence · Forecast Service · XAI Service · Scenario Service
DATA	Time-Series DB (CEB 15-min demand · Weather · Calendar) · Application DB (Labels · Reports · Logs)

AI Layer = Behaviour Intelligence (C1) · Adaptive Forecast (C2) · Explainable AI (C3) · Scenario Support (C4)

How the Modules Work Together

Data flows through all 4 AI modules to produce unified intelligence output for energy planners

📦 Data Input

- CEB 15-min demand
- Weather data
- Calendar/holidays
- 189,000+ records

🔍 C1: Behaviour

- Cluster demand days
- Detect anomalies
- Label: Normal/Peak/ Holiday/Abnormal

📈 C2: Forecasting

- CNN-LSTM + Transformer
- Short & long-term
- Confidence intervals
- Probabilistic range

🧠 C3: XAI

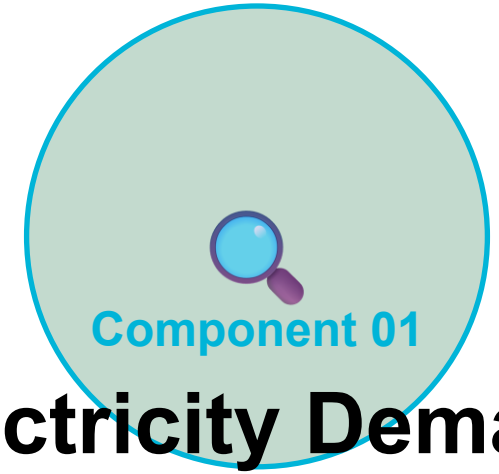
- SHAP/LIME analysis
- Factor contributions
- Cause-effect reports

⚙️ C4: Scenario & Decision Support

- User inputs: temp rise, industrial load, holidays
- Simulates what-if scenarios on demand
- Predicts adjusted demand + risk level
- Generates alerts for critical conditions

🎯 Unified Output → Energy Planners

- WHAT: demand forecast with confidence range
- WHY: causal factor explanation per day
- HOW: behaviour label (Normal/Peak/Abnormal)
- WHAT IF: scenario simulation & risk alerts



Electricity Demand Behaviour Intelligence Module

Thanujan T | IT22545176

C1: Behaviour Intelligence — Problem & Objective

Thanujan T | IT22545176 | Component 01

The Problem

✗ Existing forecasting tools only output numbers — no behaviour understanding

✗ No system classifies demand days into meaningful categories

✗ Anomalies like spikes/drops go undetected at system level

✗ No unified framework: clustering + anomaly + labels in one place

The Objective

✓ Preprocess & clean CEB 15-min demand data

✓ Generate 96-point daily demand profiles

✓ Apply K-Means & DBSCAN clustering to group similar days

✓ Detect anomalies using Isolation Forest & Z-Score

✓ Assign behaviour labels: Normal / Peak / Holiday / Abnormal

✓ Evaluate: Silhouette · Davies-Bouldin · F1-Score · Expert review

C1: Methodology — 6-Phase Pipeline

P1 Preprocessing

Missing values · Outliers · Standardize · 96-pt daily profiles

P2 Clustering

K-Means + DBSCAN · Silhouette · Elbow Method

P3 Anomaly Detection

Isolation Forest · Z-Score · Threshold calibration

P4 Behaviour Labeling

Normal Weekday · Peak Day · Holiday · Abnormal Day

P5 Visualization

Demand profile plots · Cluster charts · Calendar maps

P6 Validation

Silhouette · Davies-Bouldin · F1-Score · CEB expert review

C1: Core Novelty — Behaviour Label Taxonomy

First system to classify every demand day into human-readable categories for Sri Lanka CEB operations

Normal Weekday Demand

- Typical Mon–Fri consumption
- Morning & evening peaks present
- Baseline for all comparisons
- Majority of working days

Peak Demand Day

- Exceeds historical average
- Detected via anomaly + cluster
- Hot season / industrial surge
- Early warning → generation planning

Holiday / Low Demand

- Public holidays & weekends
- Reduced industrial load
- Flat, low-magnitude curve
- Cost optimization opportunity

Abnormal Demand Day

- Isolation Forest outlier flag
- Z-Score deviation is high
- Caused by outages/extreme events
- Triggers immediate operator alert

C1: Expected Outputs & Evaluation

Technical Outputs

- ✓ Cleaned Sri Lanka CEB demand dataset (189K+ records)
- ✓ 96-point daily demand profile generator
- ✓ Clustered demand day groups (K-Means + DBSCAN)
- ✓ Anomaly score dataset with flagged unusual days
- ✓ Complete behaviour-labeled dataset (all 4 categories)
- ✓ Interactive Behaviour Intelligence Dashboard (React.js)
- ✓ FastAPI backend with secured REST endpoints

Evaluation Metrics

Silhouette Score

Measures how well days are grouped — closer to 1.0 is better

Davies-Bouldin Index

Measures cluster separation — lower is better

Precision / Recall

How accurately anomalies are detected vs. missed

F1-Score

Balanced accuracy of anomaly detection system

Expert CEB Review

Operators validate behaviour labels match real grid events

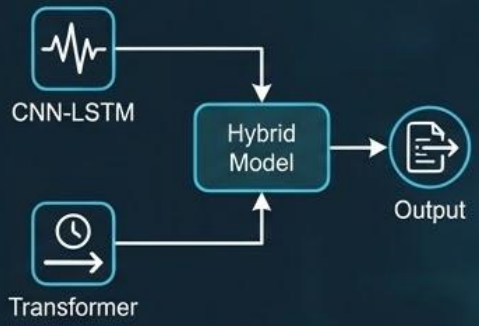
Adaptive forecasting & Prediction System

Multi-Source Data Integration



Combines historical electricity loads with weather, calendar, and economic data for a complete view of demand drivers.

Hybrid Deep Learning Models



Uses CNN-LSTM and Transformer models to capture both local patterns and long-term dependencies.

Probabilistic Forecasting



Provides confidence ranges for predictions, helping manage uncertainty and risk in energy planning.

Janahan. R IT Number: IT22069740

Research Gap

1. Data Integration Gap



Traditional models rely mostly on historical load data.

2. Advanced Deep Learning Gap



Limited use of hybrid models like CNN-LSTM or Transformer architectures.

3. Probabilistic Forecasting Gap



Most models give only point predictions, not ranges. Uncertainty quantification (e.g., confidence intervals, Monte Carlo Dropout) is rarely applied.

Technical Flow



1. Data Collection & Preprocessing

Clean and normalize data from electricity grids and weather sources.



2. Feature Engineering

Generate lagged variables, rolling statistics, and time-series embeddings.



3. Model Training

Train CNN-LSTM and Transformer models. Fine-tune hyperparameters.



4. Behavioral Analysis

Detect patterns and anomalies.



5. Probabilistic Forecasting

Compute prediction intervals to quantify uncertainty.



6. Visualization & Decision Support

Show forecasts, intervals, and cause-effect insights via dashboard.

Success Metrics & Validation for Your Model



1. Point Forecast Metrics (Traditional Accuracy Measures)

MAE (Mean Absolute Error): Average absolute difference between predicted and actual load.

Lower MAE → better prediction accuracy.

RMSE (Root Mean Squared Error): Measures large errors more heavily.

Useful to penalize big mistakes in peak demand predictions.

MAPE (Mean Absolute Percentage Error): Percentage error, easy for stakeholders to interpret.



2. Probabilistic Forecast Metrics (Uncertainty-aware Validation)

PICP (Prediction Interval Coverage Probability): Prediction ICoverage Interval)': Percentage of actual demand points falling within the predicted confidence interval.

Example: 95% PICP → 95% of actual demand values are within the model's predicted range.

Why This Output Matters

Better Planning for Utilities

Optimizes resource allocation and grid stability.

Supports Decision-Making

Provides data-driven insights for strategic choices.

Cost Savings

Reduces operational expenses and minimizes waste.

Environmental Benefits

Supports Sri Lanka's SDG goals for clean energy and climate action.

Technology Stack Python TensorFlow React Google Colab

Explainable & Causal AI System

Madushani R.P | IT Number: IT22273062



Explainable & Causal AI System

- This component explains **why electricity demand changes**.
- Traditional AI models only predict numbers but **do not explain the reasons**.
- Our system uses **Explainable AI techniques** to identify the main factors affecting electricity demand.
- It provides clear insights for **energy planners and decision-makers**.

Data Used for Analysis

The system analyzes several types of data:



Historical electricity consumption data



Weather data (temperature, rainfall)



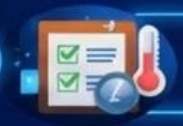
Calendar information (weekday / holiday)



Industrial electricity usage

These factors influence electricity demand patterns.

Explainable AI Process

1.  Collect **electricity demand and related** data
2.  **Train** an **AI model** (Random Forest / Gradient Boosting)
3.  **Apply Explainable AI** methods (SHAP / LIME)
4.  **Identify** factor contributions to **demand** changes
5.  **Generate explanations** and visual reports

This helps understand how different factors influence electricity demand.

Example Output

Analysis of Electricity Demand for a Specific Day

Factor	Impact on Demand
Temperature (35°C)	⬆ +50 MW
Industrial Activity	⬆ +30 MW
Holiday Effect	⬇ -10 MW



Final Electricity Demand = **950 MW**

This shows **why demand increased** on that day.

Key Contribution of the Component



- Makes AI predictions transparent and understandable



- Identifies main causes of electricity demand changes



- Detects unexpected demand spikes or drops



- Provides data-driven insights for energy management.

This improves decision-making in electricity demand management.



Research Contribution

- Introduces **Explainable AI (XAI)** for electricity demand analysis.
- **Identifies key factors** influencing demand changes such as temperature, holidays, and industrial activity.



- Provides clear **explanations** for **demand spikes** and **drops** using AI interpretation methods.

✓ **This helps stakeholders understand the reasons** behind demand variations.



Research Novelty

- Most existing systems **only** predict electricity demand values.
- Our approach focuses on **explaining** why demand changes.

★ **This** makes the system **more transparent** and **useful for energy planners**.

Scenario Analysis & Decision Support System

Dewmini D.M.D.P

IT Number:IT22630216



Scenario Analysis & Decision Support System

- This component predicts how electricity demand changes under different hypothetical conditions.
- Traditional systems show only numbers but do not test “what-if” scenarios.
- Our system allows planners to simulate changes in temperature, industrial load, or holidays.
- It provides clear insights and risk assessment to support better energy management decisions.

Key Contribution of the System



Simulates ‘what-if’ scenarios for electricity demand.



Helps planners see how changes affect future demand



Supports better decisions to prevent shortages or overloads

This aids in effective electricity demand management.

PURPOSE / INTRODUCTION



Predicts electricity demand under future hypothetical conditions.

Evaluates how changes in temperature, rainfall, industrial activity or holidays affect demand

Provides decision support to prevent over–or under-supply.

SCENARIO INPUTS



Weather conditions:
temperature, rainfall



Industrial load: factory production increase/decrease



Holidays / special events



User-defined inputs
via system interface

SIMULATION PROCESS



AI predicts electricity demand under each scenario.



Compares scenario demand with normal conditions



Detects potential spikes or drops in demand.

RISK ASSESSMENT & ALERTS



Evaluates risks: overload, under-supply, potential blackouts

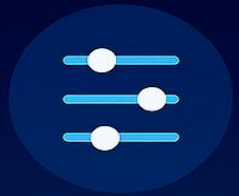


Generates alerts for planners with risk levels (green /yellow/red)



Supports proactive energy management

INTERACTIVE DECISION SUPPORT



Users adjust scenario parameters using sliders



Compare multiple scenarios side by side



Charts visualize effects for easy decision-making

BENEFITS



Enables planners to prepare for future changes



Reduces blackouts and energy waste



Supports data-driven and risk-aware decision-making